

1680. Concatenation of Consecutive Binary Numbers

Solved 

Medium

 Topics

 Companies

 Hint

Given an integer n , return the **decimal value** of the binary string formed by concatenating the binary representations of 1 to n in order, **modulo** $10^9 + 7$.

Example 1:

Input: $n = 1$

Output: 1

Explanation: "1" in binary corresponds to the decimal value 1.

Example 2:

Input: $n = 3$

Output: 27

Explanation: In binary, 1, 2, and 3 corresponds to "1", "10", and "11".

After concatenating them, we have "11011", which corresponds to the decimal value 27.

Example 3:

Input: $n = 12$

Output: 505379714

Explanation: The concatenation results in

"1101110010111011110001001101010111100".

The decimal value of that is 118505380540.

After modulo $10^9 + 7$, the result is 505379714.

Constraints:

- $1 \leq n \leq 10^5$

Python:

class Solution:

def concatenatedBinary(self, n: int) -> int:

MOD = $10^9 + 7$

res = 0

bits = 0

for i in range(1, n+1):

if (i & (i-1)) == 0:

bits += 1

res = ((res << bits) | i) % MOD

return res

JavaScript:

var concatenatedBinary = function(n) {

const MOD = 1000000007n;

let res = 0n;

let bits = 0n;

for(let i=1n; i<=BigInt(n); i++){

if((i & (i-1n)) === 0n) bits++;

res = ((res<<bits) | i) % MOD;

}

return Number(res);

```
};
```

Java:

```
class Solution {  
    public int concatenatedBinary(int n) {  
        final int MOD = 1_000_000_007;  
        long res = 0;  
        int bits = 0;  
  
        for(int i=1; i<=n; i++){  
            if((i & (i-1)) == 0) bits++;  
            res = ((res<<bits) | i) % MOD;  
        }  
        return (int)res;  
    }  
}
```